

Selected Publications: GWAS & NGS

Genome-Wide Association Studies (GWAS)

- Handedness GWAS (Meta-analysis) — Nat Hum Behav (2020)
- Basal Cell Carcinoma — PLoS One (2011)
- Nevus Count / Melanoma Risk (NID1) — Hum Mol Genet (2011, Co-first)
- Cutaneous BCC & SCC — Hum Mol Genet (2011)
- Pancreatic Adenocarcinoma Survival — Gut (2012)
- Severe Asthma Exacerbation (RF GWAS) — BMC Med Genet (2011)
- eQTL Mapping — Hum Mol Genet (2010)
- ABO & Pancreatic Cancer — Cancer Epidemiol Biomarkers Prev (2010)
- Blood Type & Breast Cancer — Int J Cancer (2011)

NGS & Functional Genomics

- Synthetic transcription elongation factors — Science (2017)
- YAP1 → USP31 → NFκB (Sarcoma) — Cancer Research (2018)
- TRIM24 degrader — Nat Chem Biol (2018)
- MYC dependency in ovarian cancer — eLife (2018)
- Super-enhancer BCL2 mechanism — Cancer Cell (2018)
- Warfarin pharmacogenetics — Pharmacogenomics (2015)

ML Expertise & Industry Experience

Functional Genomics & Machine Learning

- ENCODE Pilot Project — Nature (2007)
- Regulatory polymorphisms / eQTL — Genomics (2008)
- Lgl1 regulation during alveolarization — Pediatric Research (2010)
- RF model for asthma exacerbation — BMC Med Genet (2011)
- LSTM Stock Prediction — Kaggle Tutorial (2018)
- Biological information retrieval — ACM CIKM (1998)

Industry Highlights

- Vidagen (Founder): Genetic testing; developed dozens of products; ~ a dozen patents. (Awards: Hundred, Yellow Crane, 3551 Talent Programs).
- Booz Allen Hamilton (Sr. Data Scientist):
 - Genomic data preparation for Million Veteran Program (MVP).
 - Natural Language Processing (NLP) for NIDDK.
- D2M Biotherapeutics: Causal gene identification for cancers and immune diseases (11M+ MR results).
- Head of Biosciences (Singapore): Quantum-physics for biomedical applications.